INCH-POUND
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SUPERSEDING
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7 June 2021

FEDERAL SPECIFICATION
WASHER, FLAT (PLAIN)

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1 SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plain, flat washers.

1.2 Classification. Flat washers covered by this specification shall be of the following types, grades and classes, as specified (see 6.2).

Type:

A - Round washer
B - Square washer

Grade:

I - General assembly purpose washer
II - Aeronautical assembly purpose washer

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Class:

A - Carbon steel washer
B - Corrosion-resisting steel washer
C - Aluminum washer
D - Aluminum alloy washer
E - Brass washer
F - Copper washer
G - Nickel-copper alloy washer
H - Plastic (nylon) washer

2 APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL SPECIFICATIONS

QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip Wire, Forgings and Structural and Special Shaped Sections

FEDERAL STANDARDS

FED-STD-123	-	Marking for Shipment (Civil Agencies)

MILITARY SPECIFICATIONS

MIL-DTL-13924	-	Coating, Oxide, Black, For Ferrous Metals
MIL-DTL-16232	-	Phosphate Coating, Heavy, Manganese or Zinc Base
MIL-F-495	-	Finish, Chemical, Black, For Copper Alloys
MIL-PRF-8625	-	Anodic Coatings For Aluminum and Aluminum Alloy
MIL-PRF-32647	-	Zinc-Nickel Electroplating for Fasteners

MILITARY STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of these documents are available from https://quicksearch.dla.mil.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.21.1	-	Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

(Copies of this document are available from www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org.)

ASTM INTERNATIONAL

ASTM A153/A153M	-	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
ASTM A342/A342M	-	Standard Test Methods for Permeability of Weakly Magnetic Materials
ASTM A967/A967M	-	Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts
ASTM B36/B36M	-	Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B121/B121M	-	Standard Specification for Leaded Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B152/B152M	-	Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar
ASTM B209/B209M	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B339	-	Standard Specification for Pig Tin
ASTM B545	-	Standard Specification for Electrodeposited Coatings of Tin
ASTM B633	-	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM D3951	-	Standard Practice for Commercial Packaging
ASTM D4066	-	Standard Classification System for Nylon Injection and Extrusion Materials (PA)

(Copies of these documents are available from www.astm.org.)

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AEROSPACE INDUSTRIES ASSOCIATION/NATIONAL AEROSPACE STANDARD (AIA/NAS)

AIA/NAS NAS1149	-	Washer, Flat

(Copies of this document are available from www.aia-aerospace.org.)

SAE INTERNATIONAL

SAE AMS2700	-	Passivation of Corrosion Resistant Steels
SAE AIR4127	-	Steel: Chemical Composition and Hardenability
SAE AMS-C-81562	-	Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited)
SAE AMS-QQ-A-250	-	Aluminum and Aluminum Alloy, Plate and Sheet General Specification for
SAE AMS-QQ-N-290	-	Nickel Plating (Electrodeposited)
SAE AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)

(Copies of these documents are available from www.sae.org.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specified exemption is obtained.

3 REQUIREMENTS

3.1 Material.

3.1.1 Class A carbon steel. Carbon steel washers shall be low carbon steel in accordance with SAE AIR4127.

3.1.2 Class B corrosion-resistant steel. Corrosion-resisting steel washers shall be type 301, 302, 304, 305 or 316 in accordance with SAE AIR4127.

3.1.2.1 Magnetic permeability. Magnetic permeability shall be 2.0 maximum (air = 1.0) for a field strength of H = 200 oersteds when using a magnetic indicator per ASTM A342/A342M.

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3.1.3 Class C – aluminum. Aluminum washers shall be in accordance with ASTM B209/B209M, alloy 1100 or 3003, temper H18 or H28.

3.1.4 Class D - aluminum alloy. Aluminum alloy washers shall be in accordance with SAE AMS-QQ-A-250, alloy 2024, temper T4.

3.1.5 Class E – brass. Brass washers shall be in accordance with ASTM B36/B36M and ASTM B121/B121M, composition 11, temper half-hard.

3.1.6 Class F – copper. Copper washers shall 'be in accordance with ASTM B152/B152M, temper cold-rolled, hard.

3.1.7 Class G - nickel-copper alloy. Nickel-Copper alloy washers shall be in accordance with QQ-N-281, Class A, cold rolled, annealed.

3.1.8 Class H - plastic (nylon). Plastic (nylon) washers shall be in accordance with ASTM D4066, Type I, natural color.

3.2 Protective finish. Unless otherwise specified (see 6.2), washers shall be furnished plain, with no protective finish. When finishes are specified, they shall be in accordance with the following paragraphs.

3.2.1 Class A - carbon steel washers.

3.2.1.1 Cadmium plating (Obsolete for New Design, see 6.3). Cadmium /1 plate typing shall be in accordance with SAE AMS-QQ-P-416; Type II, Class 3 or SAE AMS-C-81562, Type II, Class 3.

/1 Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 is a suitable replacement with design authority approval.

3.2.1.2 Zinc-Nickel plating. Zinc-Nickel plating shall be in accordance with MIL-PRF-32647, Type K16.

3.2.1.3 Zinc coating. Zinc coating shall be in accordance with ASTM B633 /2, Type II or SAE AMS-C-81562, Type II. The thickness on washers for Bolt sizes thru 0.250 inch shall be Class 3 and on washers for bolt sizes over 0.250 inch shall be Class 2.

/2 Type II passivation per ASTM B633 and SAE AMS-C-81562 contains hexavalent chromium which is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc with hex free passivation per ASTM F1941/F1941M Fe/Zn 12 or Zinc-Nickel plate per MIL-PRF-32647 Type K20 are suitable replacements with design authority approval.

3.2.1.4 Galvanized zinc. Hot-dip galvanizing shall be in accordance with ASTM A153/A153M. The thickness shall be 0.00045 inch minimum.

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3.2.1.5 Phosphate coating. Phosphate coating shall be in accordance with MIL-DTL-16232, Type Z, Class 2.

3.2.2 Class B - corrosion-resisting steel washer.

3.2.2.1 Passivation. Passivation treatment shall be in accordance with ASTM A967/A967M, Nitric treatment 2-5, or Citric treatment 1-5, no post cleaning containing hexavalent chromium permitted, or with SAE AMS2700, Method 1, Types 6-8, or Method 2, no post treatment.

3.2.2.2 Black oxide. Black oxide coating shall be in accordance with MIL-DTL-13924, Class 1.

3.2.3 Classes C and D - aluminum and aluminum alloy washers.

3.2.3.1 Anodizing. Anodizing shall be in accordance with MIL-PRF-8625, Type II, Class 1. Sealers (clear or any color) containing hexavalent chromium shall be prohibited.

3.2.4 Classes E and F - Brass and copper washers.

3.2.4.1 Tin plating. Tin plating shall be in accordance with ASTM B545, ASTM B339.

3.2.4.2 Nickel plating. Nickel plating shall be in accordance with SAE AMS-QQ-N-290, Class l, Grade F.

3.2.4.3 Black chemical. Black chemical finish shall be in accordance with MIL-F-495, except hexavalent chromium shall be prohibited.

3.3 Dimensions.

3.3.1 Type A, round washers.

3.3.1.1 Grade I. Grade I, general assembly purpose washers shall be in accordance with the dimensions in the applicable military specification and ASME B18.21.1.

3.3.1.2 Grade II. Grade II, aeronautical assembly purpose washers shall be in accordance with the dimensions in AIA/NAS NAS1149.

3.3.1.3 Washer faces. Washer faces shall be parallel within 0.002 inch. Washer faces shall be flat within:

0.007 inch for bolt sizes thru 0.500 inch.
0.010 inch for bolt sizes 0.5625 thru 1.250 inches.
0.015 inch for bolts sizes over 1.250 inches.

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3.3.1.4 Concentricity. The diameter and the outside diameter shall be concentric within the tolerance of the inside diameter.

3.3.2 Type B square washers. Square washers shall be in accordance with the dimensions in Table I.

3.4 Workmanship. Washers shall be flat and smooth and free from loose scale, sharp edges and all other defects which may affect their serviceability.

Table I. Type B, grade I square washers.

Nom. bolt size	Inside diameter		Outside width		Thickness	
	Min.	Max.	Min.	Max.	Min.	Max.
.375	.438	.469	1.50	1.56	.125	1.56
.4735	.500	.531	1.75	1.81	.125	1.56
.500	.562	.594	2.00	2.06	.188	.219
.625	.719	.750	2.25	2.31	.250	.281
.750	.844	.785	2.50	2.56	.250	.281
.875	.969	1.000	3.00	3.06	.250	.281
1.000	1.094	1.125	3.50	3.56	.375	.406
1.125	1.250	1.312	4.00	4.12	.375	.406
1.250	1.375	1.429	4.50	4.62	.375	.406
1.375	1.500	1.562	5.00	5.12	.375	.406
1.500	1.625	1.688	6.00	6.12	.375	.406
1.750	1.875	1.938	6.50	6.62	.375	.406
2.000	2.125	2.188	7.00	7.12	.375	.406

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in this contract or order, the supplier may use his awn or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this and cited specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all washers of the same type, grade, class, size and protective finish, produced under the same conditions and offered for acceptance at one time.

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4.3 Sampling.

4.3.1 Sampling for examination. A random sample of washers shall be selected from each lot in accordance with ASQ Z1.4, inspection level S-2. The Acceptable Quality Level (AQL) defective shall be 4.0 percent defective.

4.3.2 Sampling for tests. The sampling method shall be in accordance with MIL- STD-1916 or ASQ H1331, table 1 or a comparable zero based sampling plan unless otherwise specified by the contract. If the applicable drawing, specification, standard, or quality assurance provision (QAP) specifies Critical, Major and/or Minor attributes, they shall be assigned Verification Levels of VII, IV and II or AQLs of 0.1, 1.0 and 4.0 respectively. Unspecified attributes shall be considered as Major unless sampling plans are specified in applicable documents. For MIL-STD-1916 the manufacturer may use the attribute or variable inspection method at their option. ASQ Z1.4 may be used to set sample lot size but acceptance is at zero non-conformances in the sample lot unless otherwise specified in the contract.

4.4 Examination. Samples selected in accordance with 4.3.1 shall visually examined for material (3.1), protective finish (3.2), dimensions (3.3), and workmanship (3.4). Corrosion resistant steel washers shall be checked for magnetic permeability (3.1.2.1). Any washer containing one more defects shall be rejected, and if the number of defective washers exceed the acceptance number, the lot represented by the sample shall be rejected.

4.5 Tests.

4.5.1 Protective finish. Tests for protective finishes shall be in accordance with the applicable specification (see 3.2).

4.5.2 Preparation for delivery. Tests for preparation for delivery shall be in accordance with ASTM D3951 or MIL-STD-2073-1 to verify conformance with Section 5.

5 PACKAGING

5.1 Packaging and packing. Packaging shall be Level A or C and packing shall be Level A, B, or C in accordance with ASTM D3951 or MIL-STD-2073-1, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military agencies. In addition to marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6 NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. Flat washers are intended for automotive, aircraft and general bolt, nut and screw application. Their purpose is to provide an increased bearing surface and to prevent galling of the base material.

6.2 Ordering data. Purchasers should select the offered options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Type, grade and class (see 1.2).
c. Protective finish, if required (see 3.2).
d. Dimensions (see 3.3)
e. Selection of applicable levels of packaging and packing (see 5.1).
f. Applicable MS or AN part number.

6.3 Military Procurement. Zinc-Nickel plated washers should be used for replacement of cadmium parts. Items procured under this specification are for military use. It is recommended the design as specified herein be evaluated for use. Cadmium or Hexavalent-Chromium should only be used in aerospace applications, as specified.

6.4 Subject term (key word) listing.

Protective finish
Cadmium
Hexavalent-Chromium
Concentricity
Zinc-Nickel

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

6.6 Supersession data. This specification includes the requirements of FF-W-92C, base document dated 7 June 2021.

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Army - AT
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Air Force - 71
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Preparing Activity:
Army - AT

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Review Activities:
Army - AV, CE
Navy - AS, MC, SH, YD

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